BrightLearn
AI & DATA SKILLS

2026 EDITION · BUILDER'S GUIDE

Data Agents

A builder's guide to designing, instructing, and deploying AI Data Agents — software that turns plain-English questions into trusted business answers in seconds.

TOPIC
Building Data Agents

SECTIONS
11 chapters

LEVEL
Beginner-friendly

INCLUDES
Definitions · Diagrams · Sample

Ask · Build · Refine
Better instructions, better insights.

00 What is a Data Agent?

Asking your data questions in plain English — and getting trustworthy answers back — is no longer science fiction. It is a skill, and a few simple ideas unlock it.

DEFINITION

Data Agent

A Data Agent is software that lets you ask questions about your business in everyday English and returns clean, trustworthy answers. It uses an LLM (like ChatGPT or Claude) to understand your question, writes the technical query against your company's data, and presents the result in plain language. Think of it as a tireless, on-demand data analyst that works at the speed of typing.

DEFINITION

Agent Instructions

Agent Instructions are the written rules that tell a Data Agent how to behave: what it is, what to do, what to avoid, and how to communicate. They are the single most important thing you write when building an agent — they turn a generic AI into *your* AI.

WITHOUT A DATA AGENT

Wait for an analyst → **Hours or days later** → **Maybe the answer**

WITH A DATA AGENT

Ask in plain English → **Agent does the work** → **Answer in seconds**

Same data, same question — but the agent removes the human bottleneck and applies your business rules every time.

01 Key Concepts You Need to Know

A few core terms come up constantly when working with Data Agents. Get comfortable with these now and the rest of this guide will make a lot more sense.

DEFINITION

Large Language Model (LLM)

An LLM is the type of AI behind tools like ChatGPT, Claude, and Gemini. It is the **brain** of a Data Agent. It reads your question, translates it into a database query, and rewrites the result in plain language. It does not 'think' like a human — it pattern-matches based on huge amounts of training data, which is exactly why your instructions matter so much.

DEFINITION

Datawarehouse

A datawarehouse is a large, well-organised database that holds your company's business data — sales, customers, products, finance. It is the **single source of truth** your Data Agent queries. Common examples include Snowflake, BigQuery, Databricks, and Microsoft Fabric.

DEFINITION

SQL

SQL (Structured Query Language) is the technical language databases speak. The **LLM writes SQL for you behind the scenes** — you never see it unless you want to. SQL is how the agent actually fetches the data needed to answer your question.

★ THE MENTAL MODEL TO REMEMBER

Imagine you have hired a brilliant new analyst on their first day. They know everything about databases and statistics in general — but they have never met you, never seen your company, and forget every conversation you have. Every time you ask them for a number, you must tell them which tables to use, what your business rules are, and what 'good' looks like. That is exactly how a Data Agent works. Your Agent Instructions are that briefing.

02 Why Data Agents Matter

Data Agents are reshaping how businesses use their data. Anyone who can build and instruct one well will work faster, ship better insights, and stand out in any data-driven role.

★ WHAT GOOD DATA AGENTS HELP YOU DO

- **Answer business questions in seconds**, instead of waiting in an analyst queue.
- **Standardise definitions** of revenue, churn, active customer — across every report and team.
- **Reduce ad-hoc requests** so your data team can focus on deeper work.
- **Catch issues early** by surfacing changes and anomalies automatically.
- **Scale data access safely** without exposing raw tables to every user.
- **Reduce wrong answers** by enforcing rules and exclusions in one place.

★ WHERE DATA AGENTS SHOW UP

- **Sales & finance** — pipeline, revenue, top accounts, period comparisons.
- **Operations** — supply chain, inventory, fulfilment metrics.
- **Marketing** — campaign performance, attribution, cohort analysis.
- **Customer service** — ticket trends, satisfaction, escalations.
- **HR & people analytics** — headcount, attrition, hiring funnels.
- **Executive dashboards** — one source of truth for the whole leadership team.

★ HOW A DATA AGENT ACTUALLY ANSWERS YOU

- **Your question** — written in plain English.
- **Your instructions** — the business rules and exclusions you have set.
- **The datawarehouse** — the actual rows it queries.
- **The LLM's reasoning** — how it translates, runs and explains the result.
- You control the first two. Those are where all the magic happens.

03 Anatomy of a Data Agent

Every Data Agent — no matter the platform — runs the same seven-step pipeline. Understand this flow once and every product on the market makes sense.

THE SEVEN-STEP PIPELINE

1. Question → **2. Understand intent** → **3. Write SQL** → **4. Query data**

5. Interpret rows → **6. Write answer** → **7. You follow up**

Steps 1 and 7 belong to you. Steps 2–6 happen inside the agent in seconds.

THE THREE INGREDIENTS

01 · THE BRAIN

LLM

The reasoning engine — ChatGPT, Claude, Gemini, or similar. Built by a vendor; you choose which one.

02 · THE TRUTH

Datawarehouse

Your business data in clean, organised tables. Already what it is — you point the agent at it.

03 · THE WISDOM

Agent Instructions

Your business rules, exclusions, definitions, tone. The only ingredient fully under your control.

04 Weak vs Strong Agent Instructions

The fastest way to feel the impact of good instructions is to see them side-by-side with weak ones. Three real-world rewrites.

✗ WEAK

"You are a data assistant. Help users with sales questions."

Why it's weak:

- No tables named
- No business rules
- Tone undefined
- No constraints on data

✓ STRONG

"You are a Revenue Insights Agent for sales leaders. Use fact_sales and dim_product. Exclude refunds and internal test accounts. Format currency as ZAR."

Why it's strong:

- Audience named
- Specific tables
- Exclusion rules listed
- Output format defined

✗ WEAK

"Rank top customers."

Why it's weak:

- By what metric?
- Top how many?
- How to break ties?
- Include all customer types?

✓ STRONG

"Rank the top 10 customers by net revenue (excluding refunds) in descending order. Break ties by unit volume, then alphabetically."

Why it's strong:

- Metric: net revenue
- N specified (10)
- Tie-breakers defined
- Exclusions stated

✗ WEAK

"Answer the user's question."

Why it's weak:

✓ STRONG

"If data is missing, state the limitation. If the question is ambiguous, state

- What if data is missing?
- What if question is unclear?
- Can it guess?
- What about zero results?

your assumption and offer to re-run. Never fabricate numbers. Treat zero rows as 'no matching records' — not zero."

Why it's strong:

- Missing-data rule
- Ambiguity rule
- No-fabrication rule
- Zero-row rule

05 The 8 Essential Instruction Sections

Strong Agent Instructions are not a paragraph — they are a structured set of eight named sections. This is the framework used by professional builders, adapted from the CitizenDeveloper365 Agent Instructions Cheat Sheet.

THE INSTRUCTION TOOLKIT

01 ROLE	02 PRIMARY OBJECTIVE	03 CORE TASKS	04 RANKING LOGIC
05 OUTPUT RULES	06 AUDIENCE & TONE	07 CONSTRAINTS	08 ERROR HANDLING

↓ COMBINE THEM ↓

A trustworthy, production-ready Data Agent

★ HOW TO USE THE NEXT THREE PAGES

The next three sections walk through all eight instruction parts in detail — grouped into Identity & Behaviour (sections 06), Logic, Format & Voice (section 07), and Safety & Honesty (section 08). Each part comes with a definition, a real example, and tag pills showing when it matters most. Read it once end-to-end; then come back to use it as a checklist when you build your own agent.

06 Identity & Behaviour Sections

The first three instruction sections answer the most fundamental questions: who is the agent, what is it for, and what does it actually do every day?

01 ROLE

Defines **who the agent is**, the domain it operates in, and the mission it is designed to accomplish. Sets the agent's identity in one paragraph.

TRY

You are a Revenue Insights Data Agent that helps sales leaders explore commercial performance. Your mission is to turn raw data into clear, decision-ready answers.

Identity	Domain	Mission	Audience

02 PRIMARY OBJECTIVE

States **the single most important outcome** the agent must deliver. Guides every downstream decision and prevents the agent from drifting into unrelated tasks.

TRY

Answer revenue and sales-performance questions accurately, applying the company's approved metric definitions. Every number must be traceable to a specific table.

Focus	Outcome	Anti-drift	Trust

03 CORE TASKS

Lists the **recurring jobs** the agent performs — the checklist of repeatable workflows. This is the agent's operating manual.

TRY

(1) Answer revenue questions from fact_sales. (2) Produce period-over-period comparisons. (3) Generate top-N rankings. (4) Surface notable changes.

Operating manual	Repeatable	Predictable

07 Logic, Format & Voice Sections

The next three sections control how the agent thinks, how it presents its answers, and how it sounds when it speaks.

04 RANKING LOGIC

Explains how the agent should **evaluate, prioritise, or classify** information. Critical for Data Agents — defines which signals matter and how to break ties.

TRY

Rank by net revenue descending. Break ties using unit volume, then alphabetical order. Never invent ranking signals not present in the source tables.

Priorities	Tie-breakers	No fabrication

05 OUTPUT RULES

Defines the **structure, formatting, and content style** of every answer. Controls headings, table formats, currency, length — anything users will see.

TRY

Structure every response as: Headline (one sentence) · Key Numbers table (≤10 rows) · Context paragraph (<60 words). Format currency as ZAR, rounded to thousands.

Structure	Format	Length

06 AUDIENCE & TONE

Describes **who the agent is speaking to and how it should sound**. A CFO and a warehouse manager need different tones even for the same question.

TRY

Your audience is sales leaders and finance partners — numerate but not *technical. Use a confident, concise business tone. Avoid SQL terminology and engineering jargon.*

Audience	Vocabulary	Tone

08 Safety & Honesty Sections

The final two sections are arguably the most important — they prevent the worst outcome of all: a confidently wrong answer that no one questions.

07 CONSTRAINTS

Specifies what the agent **must avoid** — off-limit tables, topics to refuse, assumptions never to make. These are the safety guardrails.

TRY

Use only fact_sales, fact_refunds, dim_product, dim_customer. Never query archived_sales or raw_/staging_/tmp_ tables. Exclude internal test accounts. Never provide forecasts.

Off-limits	Guardrails	Topics to refuse

08 ERROR & AMBIGUITY HANDLING

Tells the agent what to do **when data is missing, contradictory, or the question is unclear**. This is what stops a Data Agent from making things up.

TRY

If data is missing, state the limitation and offer best-effort. If a question is ambiguous, state your assumption. If a query returns zero rows, say 'no matching records' — never report it as zero.

Honesty	Assumptions	Zero-row rule

★ WHY THESE TWO SECTIONS MATTER MOST

An agent that admits 'I don't know' is far more valuable than one that confidently invents an answer. Constraints and Error Handling are what build user trust — without them, a single bad number can undo months of work. Spend extra time on these. Every edge case you describe here is one less wrong answer the business will act on.

09 Common Mistakes & Best Practices

Most 'the agent gave me a wrong answer' problems come down to a small number of recurring mistakes. Avoid these and your agent will produce trustworthy answers far more often.

⚠️ PITFALL 1 — VAGUE ROLE DEFINITIONS

Bad: "A helpful data assistant." **Better:** "A Revenue Insights Agent for sales leaders, querying fact_sales using the approved net revenue definition."

⚠️ PITFALL 2 — MISSING EXCLUSION RULES

If you do not tell the agent to **exclude refunds, internal test accounts, and archived tables**, it will not. Inflated numbers are the most common Data Agent failure mode.

⚠️ PITFALL 3 — NO OUTPUT STRUCTURE

Without a defined output format, every answer looks different. Users lose trust. **Always specify** the structure (sections, tables, currency, length) so answers are scannable and predictable.

⚠️ PITFALL 4 — NO AMBIGUITY HANDLING

When the agent has nothing to say, it will make something up — unless you tell it not to. **Always require honest 'I don't know' behaviour** for missing or unclear data.

★ BEST PRACTICES IN ONE LIST

- **Start narrow.** One audience, one domain, one table set. Expand only after the agent is reliable.
- **Name everything explicitly.** Specific table names, metric formulas, exclusion

rules.

- **Treat instructions as code.** Version them, review changes, document why each rule exists.
- **Build a regression test suite.** 10 canonical questions with known-correct answers. Run them after every change.
- **Force honesty.** A loud 'I don't know' beats a quiet wrong answer every single time.
- **Treat them as living documents.** Your business changes; your instructions must change too.

10 Data Agents in Action

Below is a complete, production-style instruction set you can copy, customise, and deploy. It uses the eight-section template for a fictional Revenue Insights Agent — your starting point for building your own.

USE CASE

Revenue Insights Agent

Helps sales leaders, finance partners, and regional managers understand commercial performance. Answers questions about revenue, top products, top customers, and period-over-period trends — using only the company's blessed datawarehouse tables.

SAMPLE INSTRUCTIONS

01 · ROLE

You are a Revenue Insights Data Agent that helps sales leaders, finance partners, and regional managers explore commercial performance using the company's datawarehouse. Your mission is to transform raw transactional data into clear, accurate, decision-ready insights that anyone in the business can act on without needing to write SQL.

02 · PRIMARY OBJECTIVE

Answer revenue, sales, and customer-performance questions accurately and quickly, applying the company's approved metric definitions and business rules. Every answer must be traceable to a specific table and metric definition, so users can trust the numbers without independent verification.

03 · CORE TASKS

First, answer revenue and sales performance questions by querying fact_sales and dim_product and aggregating using the approved revenue definition. Second, produce period-over-period comparisons (month-over-month, quarter-over-quarter, year-over-year) using dim_calendar. Third, generate top-N rankings of products, customers, and regions on request, defaulting to top 10 unless otherwise specified. Fourth, explain notable changes by surfacing the largest contributing products, customers, or regions. Fifth, always return both a short written summary and a supporting data table.

04 · RANKING LOGIC

When ranking items, default to ranking by net revenue (gross revenue minus refunds and cancellations) in descending order. Break ties using unit volume, then alphabetical order. When the user asks for "top performers" without specifying a metric, use net revenue. When the user asks about "growth," rank by absolute change in revenue period-over-period unless they explicitly ask for percentage. Never invent ranking signals not present in the source tables.

05 · OUTPUT RULES

Structure every response in three sections: a one-sentence Headline that states the answer directly, a Key Numbers table with no more than ten rows and clear column headers, and a short Context paragraph (under 60 words) that explains any notable patterns. Format all currency in South African Rand (ZAR), rounded to the nearest thousand, with a leading R symbol. Format percentages to one decimal place. Always state the date range the answer covers. Keep total response length under 250 words unless the user asks for detail.

06 · AUDIENCE & TONE

Your primary audience is sales leaders, finance partners, and regional managers — business professionals who are numerate but not technical. Use a confident, concise, business-professional tone. Avoid SQL terminology, data engineering jargon, and references to internal system mechanics. Speak in terms of business outcomes (revenue, customers, products, regions), not data structures (tables, joins, queries).

07 · CONSTRAINTS

Use only the following blessed tables: fact_sales, fact_refunds, dim_product, dim_customer, dim_region, dim_calendar. Never query the archived_sales table or any table prefixed with raw_, staging_, or tmp_. Never include internal test accounts (customer_type = 'INTERNAL_TEST') in any aggregate. Do not provide forecasts, projections, or predictions — only report on what has happened. Do not provide financial, legal, or contractual advice. Do not expose individual customer names in any aggregated answer; only show customer names when the user is querying a single, specific customer they have named themselves. Never reference internal system logic, AI mechanics, or the SQL queries you generate.

08 · ERROR & AMBIGUITY HANDLING

When required data is missing or incomplete, state the limitation clearly and provide a best-effort partial answer rather than guessing. When the user's question is ambiguous (for example, they ask about "last quarter" near a quarter boundary), state the assumption you are making and offer to re-run with a different interpretation. If a requested calculation cannot be performed with the available tables, explain why and suggest the closest available metric. If a query returns zero rows, do not report a value of zero as if it were a real result — explicitly say that no matching records were found and suggest checking filters or date ranges. Never fabricate numbers, never invent customer or product names, and never present a number with more confidence than the underlying data supports.

Better instructions.
Better insights.

Data Agents are not just for engineers — they are the new way every team will answer business questions. Practice the eight-section template on a real use case and you will feel the difference immediately.

★ BRIGHTLEARN · DATA AGENTS ★